Chapter 10 Problem Set

Siddharth K

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1. Primes less than 100 of the form a^2+ab+b^2 are $3, 7, 13, 19, 31, 37, 43, 61, 67, 73, 79,$ and 97. Based on this data, it seems like they are all of the form $6k+1$ with 3 as an exception.

2. We are given that $a^2 + b^2 = 104309$, so $29(a^2 + b^2) = 3024961 = 519^2 + 1660^2$. We know that $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$. Since $29 = 2^2+5^2$, we can say that $(2^2+5^2)(a^2+b^2) = (2a-5b)^2+(2b+5a)^2 = 519^2+1660^2$. Solving the system of equations $2a-5b = 519$ and $5a+2b = 1660$, we get $a = 322$ and $b = 25$.

3. Since $p \equiv 1 \pmod 4$, we can write $a^2 + b^2 = p^2$ for integers a and b. If a and b weren't relatively prime, then there would exist d such that $a = dx$ and $b = dy$, so $d^2(a^2 + b^2) = p^2$. That is only possible when $d = 1$ or $d = p$. The first case means a and b are relatively prime, and the second case means that $a^2 + b^2 = 1$, which means either a or b is 1 and the other is 0, which we know is impossible. Therefore, a and b are relatively prime.

4. The pair $(6, 26)$ is a counterexample. We see that $6 = 1^2 + 1^2 + 2^2$ and $26 = 1^2+3^2+4^2$, and $6\times26 = 156$ cannot be written as the sum of three squares.

5. The list of numbers up to 100 that can be written as a sum of three squares includes all numbers from 0 to 100 except:

$$7, 15, 23, 28, 31, 39, 47, 55, 60, 63, 71, 79, 87, 91, 92, 95$$

From these numbers, it appears that powers of 4 are removed, numbers congruent to $-1 \pmod 8$ are removed, and products of such numbers and powers of 4 are removed.

6. We are given that $n = \left(\frac{a}{b}\right)^2 + \left(\frac{c}{d}\right)$. Clearing denominators, we get $b^2d^2n = a^2 + b^2$. Since b^2d^2n is composite and can be written as the sum of two squares, all of its prime factors $p \equiv 3 \pmod 4$ in the prime factorization have an even exponent.

Every prime number in the prime factorization of n is congruent to either 1, 2, or $3 \pmod 4$. We already know that the primes congruent to $1 \pmod 4$

can be written as the sum of two squares. Also, the only prime congruent to 2 (mod 4) is 2, which is 1^2+1^2. Finally, we just showed that all primes congruent to 3 (mod 4) have an even power in its prime factorization. From that, we can deduce that it can be written as the sum of two squares. Since n is a product of multiple sums of two squares, it too is a sum of two squares.

7. Since $p \equiv 1 \pmod 4$, we can say that $p = 4k+1$. Therefore, $(p-1)! = (4k)! = (1 \cdot 2 \cdot \ldots \cdot 2k)((2k+1)(2k+2)\ldots(4k))$.

Let's say $a = \{1, 2, \ldots, 2k\}$. We see that $a_i(p-a_i) = a_ip - a_i^2 \equiv -a_i^2 \pmod p$ for all $i \in \{1, 2, \ldots, 2k\}$. By Wilson's Theorem, $(p-1)! \equiv -1 \pmod p$, so $\prod_{i=1}^{2k} a_i(p-a_i) \equiv -1 \pmod p$. Therefore, $\prod_{i=1}^{2k} -a_i^2 \equiv \left(\prod_{i=1}^{2k} a_i\right)^2 \equiv (2k)! \equiv \left(\frac{p-1}{2}\right)! \equiv -1 \pmod p$. We can let $n = \left(\frac{p-1}{2}\right)!$, and that would guarantee that $n^2 \equiv -1 \pmod p$.

8. Let's consider the set $S = \{1, 2, \ldots, p-1\}$. We can split that set into subsets of the form $\{x, \overline{x}, -x, -\overline{x}, ix, -ix, i\overline{x}, -i\overline{x}\}$. For all $x \in S$, there exists $\overline{x}$ because p is prime and all elements of S are smaller than p.

Those sets would all have 8 elements unless: $x = -x$, $x = \overline{x}$, $x = -\overline{x}$, $x = ix$, $x = -ix$, $x = i\overline{x}$, or $x = -i\overline{x}$. If $x = -x$, then $2x \equiv 0 \pmod p$. Since p is prime, x must be 0, which is not an element of S. If $x = \overline{x}$, then $x^2 \equiv 1 \pmod p$, which means $x = 1$ or $x = p-1$. In that case, the set would look like $\{x, -x, ix, -ix\}$, with four elements. If $x \equiv -\overline{x} \pmod p$, then $x^2 \equiv -1 \pmod p$. So the set would be $\{x, -x, ix, -ix\}$, which has four elements. If $x \equiv ix \pmod p$, then $x^2 \equiv -x^2 \pmod p$, or $2x^2 \equiv x^2 \equiv 0 \pmod p$, which cannot be possible given the elements of S. If $x \equiv -ix \pmod p$, the same result follows. If $x \equiv i\overline{x} \pmod p$, then $x^4 \equiv 1 \pmod p$. In this case, the subset would be $\{x, -x, \overline{x}, -\overline{x}\}$. The same follows when $x \equiv -i\overline{x} \pmod p$.

We know that the number of elements in S is a multiple of 8. Most subsets that we made have 8 elements except one with 4. Also, if $x^4 \equiv 1 \pmod p$, then there would be two subsets with four elements each. That means $x^2 + 1 \equiv 0 \pmod p$, so that the number of elements in S becomes a multiple of 8. Therefore, if $p \equiv 1 \pmod 8$, then there exists x such that x^2+1 is a multiple of p.

From there, I couldn't find a way to prove that $p = x^2 + 2y^2$.

9. Let's assume $p = a^2+b^2 = c^2+d^2$. We can factor that as $p = (a+bi)(a-bi) = (c+di)(c-di)$. We know that $|a+bi| = |c+di| = |a-bi| = |c-di|$ because $a^2+b^2 = c^2+d^2$. I don't know exactly how to proceed from there.